Program 6:

Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation

Step 1: create directory using command **mkdir MyGradleApp** and enter the directory using the command **cd MyGradleApp**

```
shreya@ShreyaM:~$ cd devops
shreya@ShreyaM:~/devops$ mkdir MyGradleApp
shreya@ShreyaM:~/devops$ cd MyGradleApp
```

Step 2: Type the command **gradle init –type java-application**

```
shreya@ShreyaM:~/devops/MyGradleApp$ gradle init --type java-application
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 51s
2 actionable tasks: 2 executed
shreya@ShreyaM:~/devops/MyGradleApp$
```

Step 3: Go to app directory to find build.gradle. Open build.gradle and save

```
/*
 * This build file was generated by the Gradle 'init' task.
 *
 * This generated file contains a sample Java project to get you started.
 * For more details take a look at the Java Quickstart chapter in the Gradle
 * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial_java_projects.html
 */

// Apply the java plugin to add support for Java
apply plugin: 'java'

// Apply the application plugin to add support for building an application
apply plugin: 'application'

// In this section you declare where to find the dependencies of your project
repositories {
    // Use jcenter for resolving your dependencies.
    // You can declare any Maven/Ivy/file repository here.
    jcenter()
}

dependencies {
    // This dependency is found on compile classpath of this component and consumers.
    compile 'com.google.guava:guava:23.0'

    // Use JUnit test framework
    testCompile 'junit:junit:4.12'
}

// Define the main class for the application
mainClassName = 'App'
```

Step 4: type the command **gradle build** to compile the source code and package the app.

```
shreya@ShreyaM:~/devops/MyGradleApp$ gedit build.gradle
shreya@ShreyaM:~/devops/MyGradleApp$ gradle build
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.pom
Download https://jcenter.bintray.com/com/google/guava/guava-parent/23.0/guava-parent-23.0.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_parent/2.0.18/error_prone_parent-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.pom
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-parent/1.14/animal-sniffer-parent-1.14.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/mojo-parent/34/mojo-parent-34.pom
Download https://jcenter.bintray.com/org/codehaus/codehaus-parent/4/codehaus-parent-4.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.jar
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.jar
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.jar
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.jar
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.jar

BUILD SUCCESSFUL in 57s
7 actionable tasks: 7 executed
```

Step 5: type the command **gradle run** to execute the default Hello World application

```
shreya@ShreyaM:~/devops/MyGradleApp$ gradle run

> Task :run
Hello world.


BUILD SUCCESSFUL in 10s
2 actionable tasks: 1 executed, 1 up-to-date
shreya@ShreyaM:~/devops/MyGradleApp$
```

Step 6: Go to app directory to find build.gradle. Open build.gradle and type the code in build.gradle

```
/*
 * This build file was generated by the Gradle 'init' task.
 *
 * This generated file contains a sample Java project to get you started.
 * For more details take a look at the Java Quickstart chapter in the Gradle
 * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial_java_projects.html
 */

// Apply the java plugin to add support for Java
apply plugin: 'java'

// Apply the application plugin to add support for building an application
apply plugin: 'application'

// In this section you declare where to find the dependencies of your project
repositories {
    // Use jcenter for resolving your dependencies.
    // You can declare any Maven/Ivy/file repository here.
    jcenter()
}

dependencies {
    // This dependency is found on compile classpath of this component and consumers.
    compile 'com.google.guava:guava:23.0'

    // Use JUnit test framework
    testCompile 'junit:junit:4.12'
}

// Define the main class for the application
mainClassName = 'App'

task display{
doLast{
println '6D devops class'
}
}
```

Step 7: type the command **gradle <task-name>** to run the task

```
shreya@ShreyaM:~/devops/MyGradleApp$ gradle display

> Task :display
6D devops class


BUILD SUCCESSFUL in 1m 16s
1 actionable task: 1 executed
shreya@ShreyaM:~/devops/MyGradleApp$
```

Continuous Integration (CI) Pipeline with Jenkins

Gradle with Jenkins

Step 1: in the terminal type the code MyGradleApp to enter the folder .

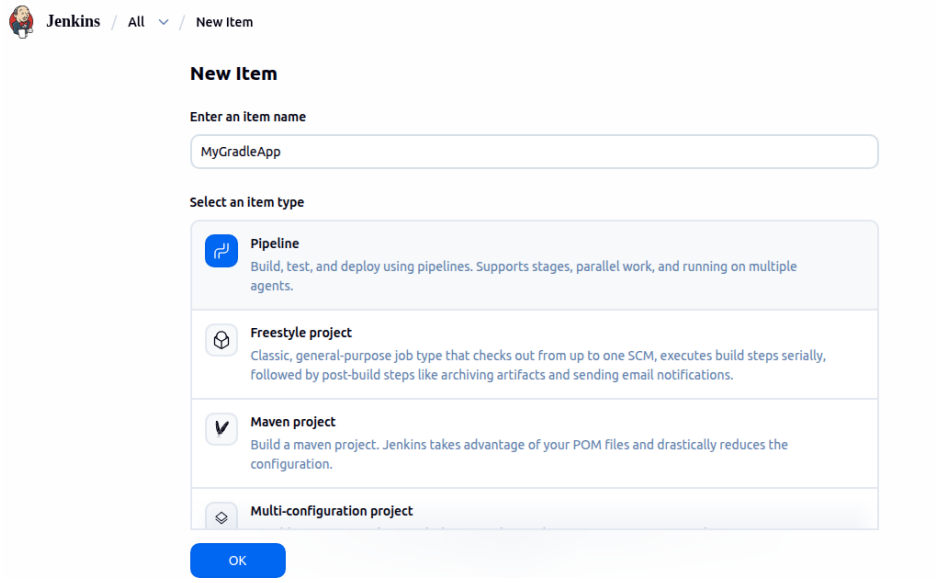
Type the command gedit JenkinsFile to open the Jenkinsfile and paste the code

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/shreya20m/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
```

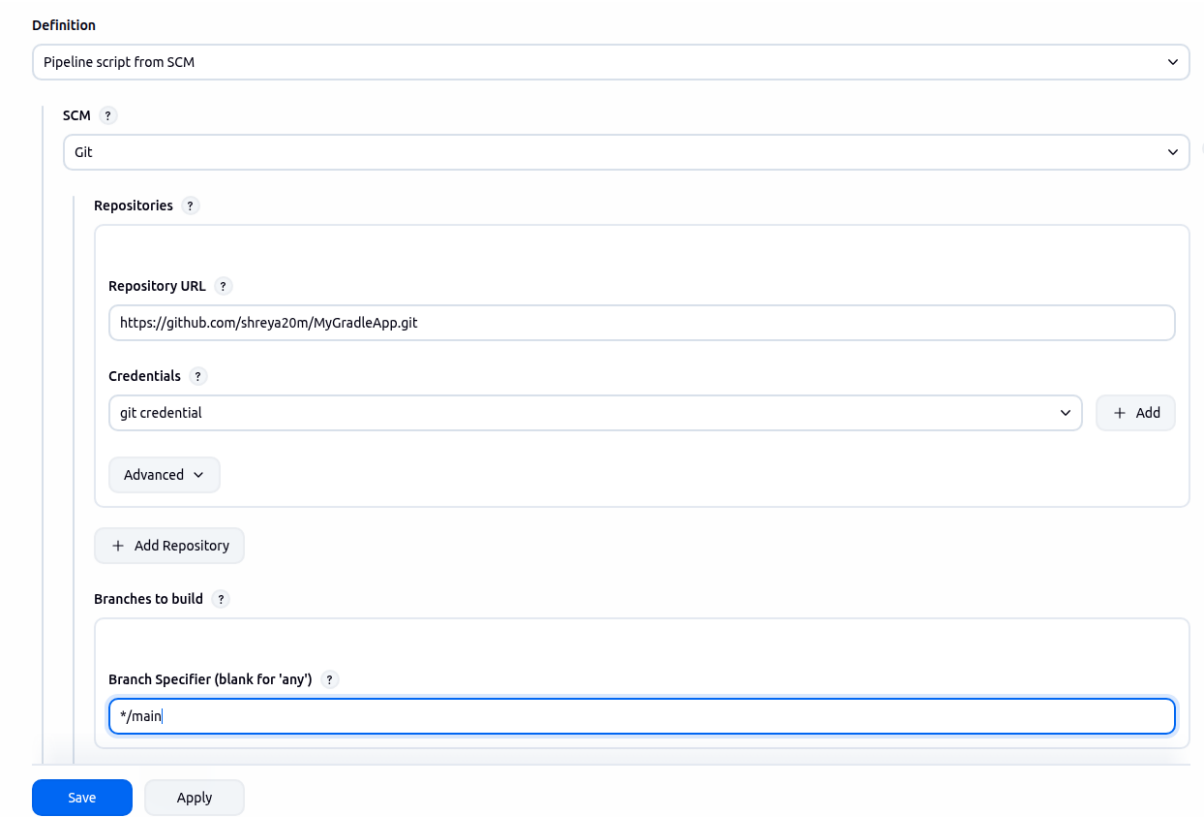
```
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }



    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 2: in Jenkins Dasboard click on New Item

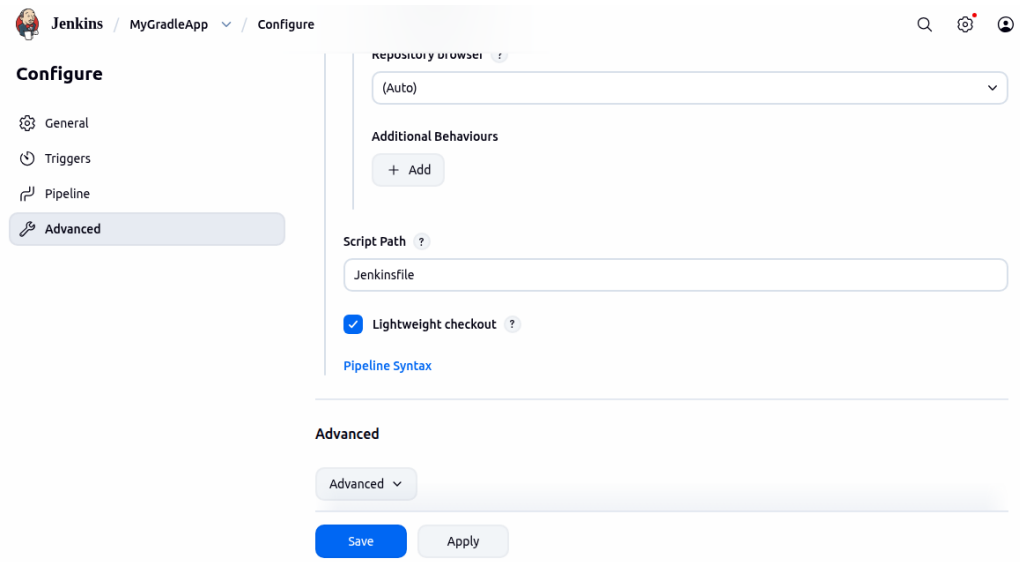
Give the name MyGradleApp and select Pipeline



In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credentials

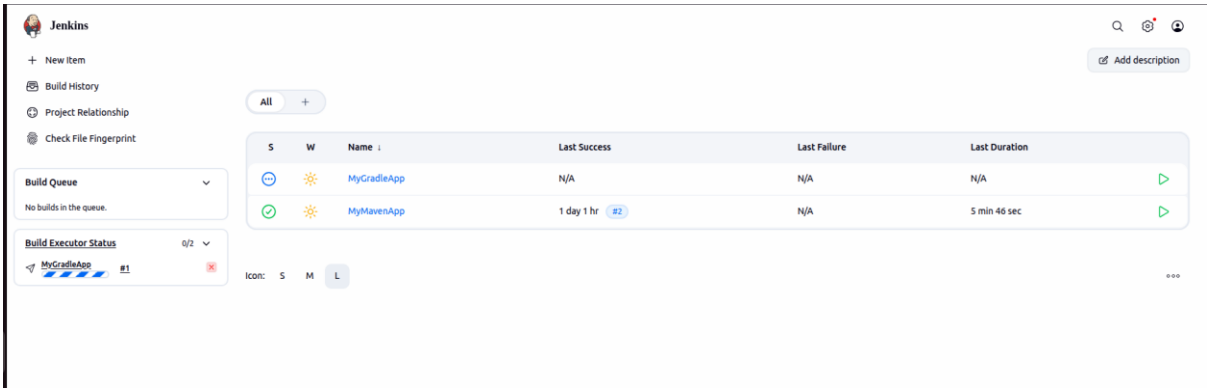
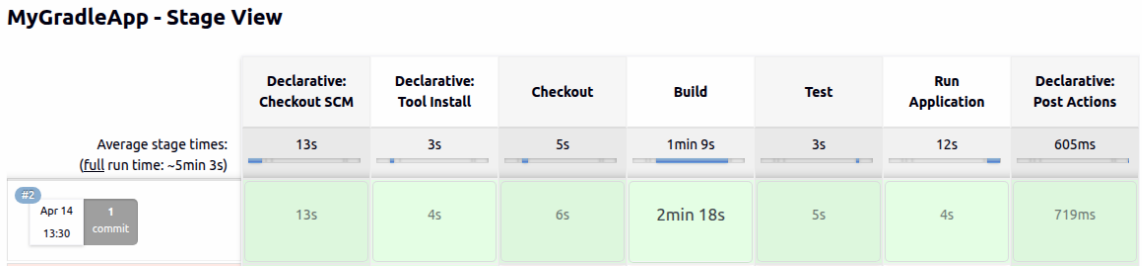


Then click on apply and save



Step 3: go to Jenkins Dasboard select the MyGradleApp pipeline and select Build Now. After build select Full Stage View

Devops



Once the errors are corrected Build again